

A gas-transport model predicts the *Arabidopsis* spaceflight transcriptome, and illumination separates six flight experiments

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Abstract

Plants flown on the International Space Station are grown inside enclosures, and the atmosphere inside those enclosures is rarely measured. We coupled a validated lattice-Boltzmann model of leaf boundary-layer gas transport to a Farquhar–von Caemmerer–Berry model of C_3 photosynthesis and asked what an illuminated, sealed canister does to the CO_2 reaching Rubisco. The model predicts that photosynthesis draws a sealed Biological Research in Canisters (BRIC) enclosure down to near its CO_2 compensation point within minutes, raising the Rubisco oxygenation fraction from 28 % to 54 % and reducing 12 h carbon gain to 1 % of an unenclosed control. Because that drawdown is a mass balance, it occurs in the ground control as well as in flight. It therefore does not bias the flight-versus-ground contrast every such experiment reports — it changes what that contrast is *about*. We estimate the enclosure effect on metabolite pools to be 25 times the microgravity effect, which means the spaceflight response is being measured in plants their own hardware has already carbon-starved. Contrary to our own starting hypothesis, the microgravity boundary-layer penalty does not amplify as CO_2 falls; it converges, leaving a persistent 5–9 % assimilation deficit. Fixing these predictions before examining any data, we tested them against OSD-522 (BRIC-LED), where the model called eight of nine gene-set responses correctly, and against five further OSDR studies. Illumination separated the six studies perfectly — every illuminated enclosure showed a carbon-starvation signature and every dark one its inverse — while the predicted gradient across enclosure types did not hold. NASA’s Open Science Data Repository contains no plant metabolome at all (0 of 66 plant studies), so the third omic layer for pathway integration had to be predicted rather than measured. We argue that enclosure atmosphere is a first-order, currently unmeasured covariate in plant spaceflight biology — not a bias to correct for, but a baseline state to measure — and set out the three-arm design and the in-canister measurement that would establish it.

Keywords: spaceflight, microgravity, photorespiration, *Arabidopsis*, growth hardware, multi-omics

1 Introduction

Interpreting a plant spaceflight experiment means attributing a molecular difference to microgravity. That attribution rests on an assumption that is almost never tested: that the flight and ground arms differ in gravity and in nothing else that matters to metabolism.

Plants flown on the International Space Station are grown inside enclosures — sealed Petri Dish Fixation Units in BRIC canisters, micropore-taped dishes in CARA, actively ventilated chambers in VEGGIE and the Advanced Plant Habitat. These enclosures differ by orders of magnitude in how freely they exchange gas with the cabin, and to our knowledge no BRIC-class flight has logged the CO_2 concentration its plants actually experienced.

The quantity that matters is not the cabin atmosphere but the CO_2 mole fraction reaching Rubisco, C_c , which sits at the end of a chain of conductances: from bulk air across the leaf boundary layer, through the stomata, and through the mesophyll. Only the first of those is gravity-dependent, through buoyancy-driven convection. Rubisco partitions between carboxylation and oxygenation as $V_o/V_c = 2\Gamma^*/C_c$ [1], so anything that lowers C_c raises photorespiration.

Here we ask a question about the enclosure rather than about gravity, and we ask it of the model before we ask it of any data.

2 Results

2.1 An illuminated sealed canister is a CO_2 -starvation chamber

Gas transport was taken from LunarLeaf-CFD [2], a lattice-Boltzmann solver validated against four analytical and empirical gates. Its enclosure mass balance draws a lit BRIC canister from 400 ppm to near zero in approximately seven minutes, and puts 12 h carbon gain at 1 % of an unenclosed control against 90 % for micropore tape and 100 % when ventilated (Fig. 1).

The step that matters for interpretation is that this drawdown is a mass balance between photosynthetic uptake and enclosure volume. It is therefore essentially gravity-independent, and it happens in the ground control as much as in flight.

2.2 What that does to Rubisco, and what gravity does not do

Feeding the boundary-layer conductance into a Farquhar–von Caemmerer–Berry model [1] with in-vivo Rubisco kinetics [3, 4] moves the leaf along its CO_2 response curve: oxygenation fraction rises from 28.3 % in ventilated hardware at ambient CO_2 to 53.7 % in a drawn-down canister (Fig. 2a, Table 1).

We began from the hypothesis that CO₂ starvation would *amplify* the microgravity boundary-layer penalty. The model says the opposite (Fig. 2b). As assimilation falls toward the compensation point, the flux through the boundary layer falls with it, so the CO₂ drop across that layer, A/g_{bl} , tends to zero and the two gravities converge. What survives is a persistent proportional assimilation deficit of about 5% at leaf scale, 7% at rosette scale and 9% in a dense canopy, nearly independent of where in the drawdown the plant sits (Fig. 2c). That insensitivity is the prediction’s strength: it does not require knowing the canister CO₂, which nobody measured.

Propagating both contrasts to metabolite pools makes the asymmetry explicit. The sealed-versus-ventilated hardware effect is 25 times the flight-versus-ground effect (Fig. 2d) — and because it acts on both arms, it cancels.

2.3 A metabolite layer that does not exist

Pathway-level integration needs a metabolome. We queried all 567 OSDR studies and cross-tabulated organism against assay type: six are metabolite profiling, and every one is mouse, human, rat or microbial. **None of the 66 plant studies has a metabolome.**

We therefore predicted one (Fig. 3), propagating the modelled fluxes to 21 compounds under one stated assumption — that at quasi-steady state a pool tracks the flux feeding it. Every value is labelled as model output in every file. Compounds the model cannot speak to, including all tricarboxylic-acid intermediates and all fermentation products, are excluded with the reason recorded rather than filled in.

2.4 The model called eight of nine gene sets, blind

OSD-522 [5] grew *Arabidopsis* in BRIC-LED hardware under 60 $\mu\text{mol m}^{-2} \text{s}^{-1}$ of red and blue light, with six flight and six ground-control shoot samples. We computed differential expression with PyDESeq2 [6], obtaining 1,795 genes at FDR < 0.05 from 20,983 tested, and used the deposited proteomics ratios for 5,160 proteins (Fig. 4).

Scoring these against KEGG-derived gene sets (Fig. 5), the model’s predictions held for eight of nine sets: the photosystem and light-harvesting apparatus fell (-0.238 , $p = 3.8 \times 10^{-6}$; and in the proteome -0.820 , $p = 5.0 \times 10^{-4}$), Rubisco fell (-0.403 , $p = 4.5 \times 10^{-3}$), carbon fixation and starch/sucrose metabolism fell, carbon-starvation DIN markers rose ($+0.686$, $p = 2.6 \times 10^{-3}$), and photorespiratory enzymes were not induced. The one mismatch is the whole KEGG glyoxylate and dicarboxylate map, which contains a good deal besides the C₂ cycle.

The sharper test is between sets, which is immune to the global depression of metabolic transcripts: photorespiratory enzymes sat above Rubisco in the transcriptome ($+0.239$, $p = 6.8 \times 10^{-3}$) and independently in the proteome ($+0.285$, $p = 4.9 \times 10^{-2}$).

2.5 Pathway integration ranks photorespiration last of 231

Submitting all three layers to PaintOmics [7] against KEGG and MapMan (Fig. 6), 98–100% of genes, 97% of proteins and all 21 predicted compounds mapped. Of

231 pathways tested, 31 reached combined Fisher $p < 0.05$, including photosynthesis ($p = 0.008$), photosynthesis antenna proteins ($p = 0.022$) and starch and sucrose metabolism ($p = 0.019$). Glyoxylate and dicarboxylate metabolism — the photorespiration map — ranked *last of all 231* at $p = 0.994$.

The metabolite hub analysis is the least circular result available here, because the compounds are predicted but the differentially expressed genes counted around them are measured. Seven of fifteen compounds reached $\text{FDR} < 0.05$, and every one was a carbon-starvation or carbon-supply node. Not one of the eight C_2 photorespiratory intermediates was a significant hub, although all eight mapped.

2.6 Illumination separates six studies; the enclosure gradient does not

The model’s claim is an ordering, so we tested it across six matched flight-versus-ground contrasts spanning three enclosure classes [8–11] (Fig. 7; selection and exclusions in Table 2).

Illumination separated them completely. All three illuminated studies showed the starvation-over-photosynthesis signature (+0.41, +0.92, +2.58) and all three dark studies its inverse (−0.39, −1.95, −3.22); one-sided Mann–Whitney $p = 0.05$, the floor attainable at three versus three. The cleanest evidence is internal: OSD-678 light (+2.58) against OSD-678 dark (−0.39) is the same hardware, genotype and flight.

The predicted gradient across enclosure classes did *not* hold (Spearman $\rho = -0.50$, $p = 0.67$). Sealed and ventilated fell the right way round but micropore tape sat highest, and three studies cannot rank three classes. We report this as a failure of that prediction. The driver appears to be illumination inside an enclosure rather than sealing as such — which is mechanistically what the model says, since drawdown requires photosynthesis to be running.

3 Discussion

The strongest signal in the OSD-522 integration is one this model does not explain: protein processing in the endoplasmic reticulum, at $p = 8.6 \times 10^{-9}$, the unfolded protein response. A gas-transport account explains part of this experiment, not all of it.

What the analysis does support is narrower and, we think, more useful. In an illuminated enclosure the atmosphere the plants experience is set by the hardware, it is not measured, and it is large. Because it acts on both arms it does not bias the published contrast — which is precisely why it has gone unnoticed. What it does instead is set the physiological state in which the microgravity response is read out: not a healthy plant, but a carbon-starved one. That is a baseline condition, not a bias to correct for, and a sensor would settle it.

3.1 Limitations

Cross-study comparison is the weakest joint here: different laboratories, durations, genotypes and harvest protocols. We restricted to GeneLab-standard RNA-seq, above-ground tissue and matched contrasts, and we report gene-set *ordering* rather than absolute effect sizes, but the limitation stands. The illumination result rests on three studies per group. The intended ventilated proteome arm does not exist, because OSD-16 has no processed files, so the two-layer comparison is confined to sealed hardware. The metabolite layer is predicted throughout, and the metabolite class-activity test is close to circular because it tests our own predicted list; the hub analysis is not, and is reported as the informative one. Finally, V_{cmax} , J_{max} , R_d , g_s and g_m remain literature values for *Arabidopsis* rather than fits to flight material, and g_s is held fixed although BRIC canisters are humid.

3.2 What would settle it

Three things, in increasing cost. First, log CO₂ and O₂ inside the canister: the central assumption of this analysis becomes a measurement, and an entire archive of BRIC-derived transcriptomes becomes reinterpretable. Second, run a targeted metabolite panel on archived flight material — roughly twenty compounds would be the first plant spaceflight metabolome in existence, and our predicted set is a ready-made target list. Third, fly a three-arm design (Fig. 8): flight sealed, ground sealed, ground ventilated. Two arms confound hardware with gravity; three separate them, and the third arm never leaves the ground.

4 Methods

Full methods, every script and every derived table are in the repository accompanying this manuscript. Study selection, including each exclusion and its reason, is in `data/study_registry.tsv`.

Gas transport. Boundary-layer conductance and enclosure mass balances were taken from LunarLeaf-CFD [2] and vendored into this repository.

Photosynthesis model. A Farquhar-von Caemmerer-Berry C₃ model [1] with the oxygenation term explicit, using in-vivo temperature responses [3, 4], solved for the supply-equals-demand operating point.

Differential expression. OSD-522 was processed with PyDESeq2 [6]. The other five studies use GeneLab’s own deposited differential-expression tables, with matched flight-versus-ground contrasts selected by pattern and recorded in the registry.

Gene sets. Built from KEGG pathway membership and description regular expressions fetched live [12], never typed from memory.

Pathway integration. PaintOmics [7], organism `ath`, KEGG and MapMan, with the optional AI interpretation disabled. PaintOmics reports enrichment but not the gene lists behind it, and MapMan has no equivalent of the KEGG REST API, so the gene membership used in the accompanying data explorer was reconstructed from GoMapMan’s Araport11 gene-to-bin mapping and the MapMan diagram layouts, by PaintOmics’ own inclusion rules. It reproduces PaintOmics’ feature count exactly or

Table 1 Modelled operating points. Flight and ground control sit at effectively the same place because the sealed canister draws both arms down; the ventilated arm does not.

Arm	g_{bl}	C_a	C_c	φ (%)	A
Ground control (sealed)	0.540	100	75.5	53.1	2.40
Flight (sealed)	0.291	100	73.9	53.7	2.22
Ventilated (ambient)	0.291	400	216.9	28.3	15.56

g_{bl} in $\text{mol m}^{-2} \text{s}^{-1}$; C_a and C_c in $\mu\text{mol mol}^{-1}$; A in $\mu\text{mol m}^{-2} \text{s}^{-1}$. Rosette scale.

Table 2 The six studies analysed, and the six excluded. Tissue is the binding constraint: a photosynthesis prediction says nothing about roots.

Study	Hardware	Class	Tissue	Light
OSD-522	BRIC-LED	sealed	shoots	lit
OSD-38	BRIC-PDFU	sealed	whole seedlings	dark
OSD-321	BRIC-PDFU	sealed	seedlings	dark
OSD-678	CARA	tape	leaves	lit and dark
OSD-427	VEGGIE	vented	leaf	lit

Excluded: OSD-120, OSD-193, OSD-281 (roots); OSD-218 (tissue unstated); OSD-16 (no processed files); OSD-7 (microarray, not comparable).

within three features for 9 of the 13 significant MapMan diagrams and runs larger for the four whole-ontology overview maps; both counts are reported rather than one, and the discrepancy on those four is not explained by any rule recoverable from PaintOmics’ published sources.

Supplementary information. Supplementary Figure S1 places every locus in the significant pathways into its transcript and protein differential-expression bin, and reports how far the reconstructed MapMan membership agrees with PaintOmics’ own feature counts.

Data availability. All primary data are public in the NASA Open Science Data Repository [13]: OSD-522, OSD-38, OSD-321, OSD-678 and OSD-427. Derived tables and figures are archived with this manuscript. TODO insert the Zenodo DOI on deposit.

Code availability. TODO insert the repository URL and archived release DOI.

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Competing interests. The authors declare no competing interests.

Growth hardware, not gravity, sets the atmosphere at the leaf

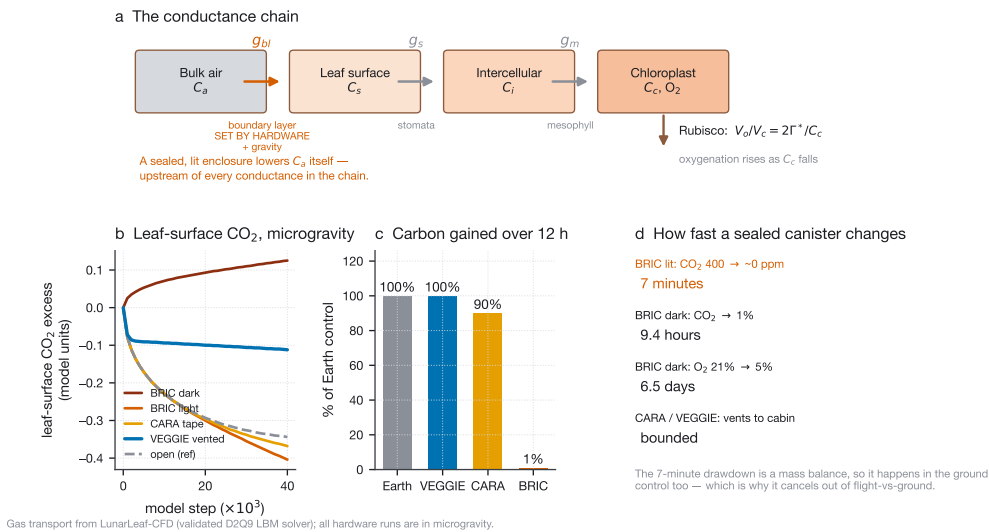
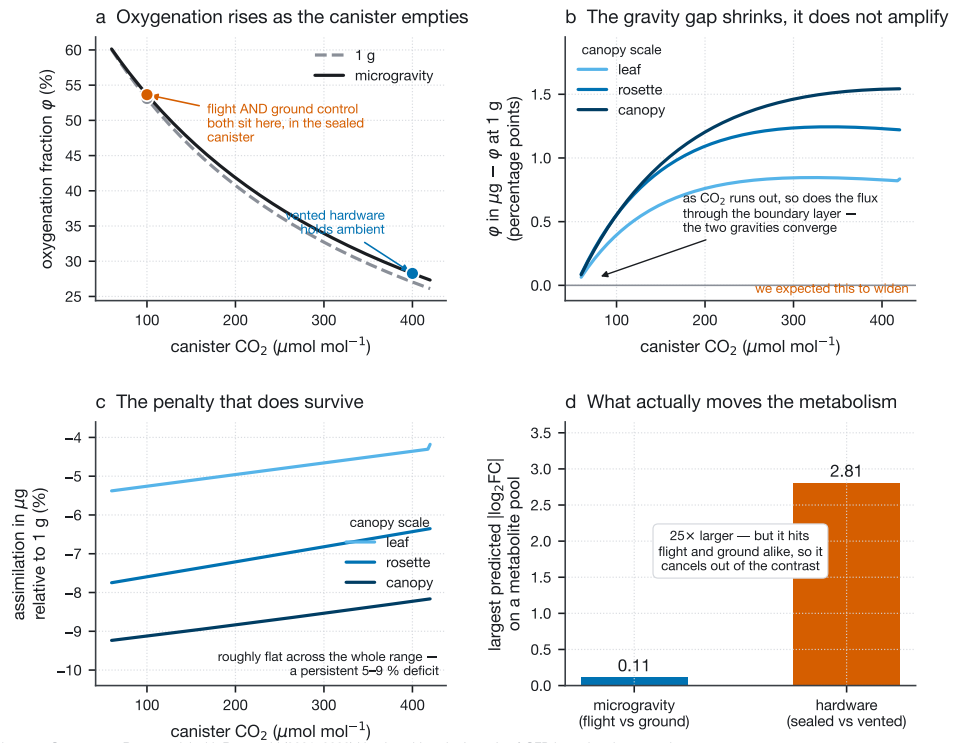


Fig. 1 Growth hardware, not gravity, sets the atmosphere at the leaf. **b** shows CO₂ at the leaf surface, which is defined for all five cases and is the quantity the photosynthesis model consumes; the alternative series in the solver output, the mean over the enclosure volume, exists only for the three closed cases, because a vented enclosure has no closed volume to average.

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A drawn-down enclosure, not microgravity, is what reaches Rubisco



Farquhar-von Caemmerer-Berry model with Bernacchi (2001, 2003) kinetics, driven by LunarLeaf-CFD boundary-layer conductance.

Fig. 2 A drawn-down enclosure, not microgravity, is what reaches Rubisco. Panel b shows the refutation of our own starting hypothesis.

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The metabolite layer had to be predicted — OSDR holds no plant metabolome

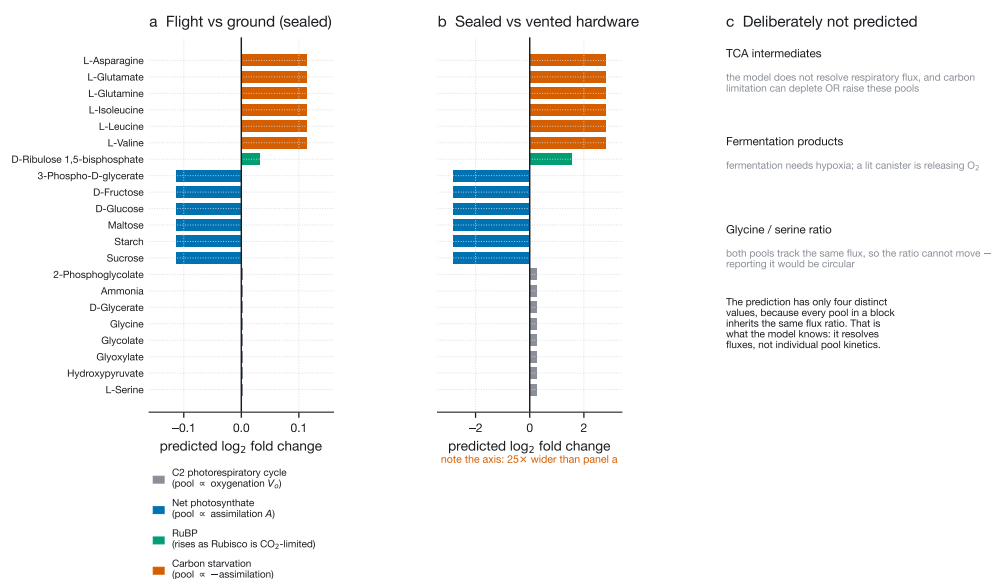
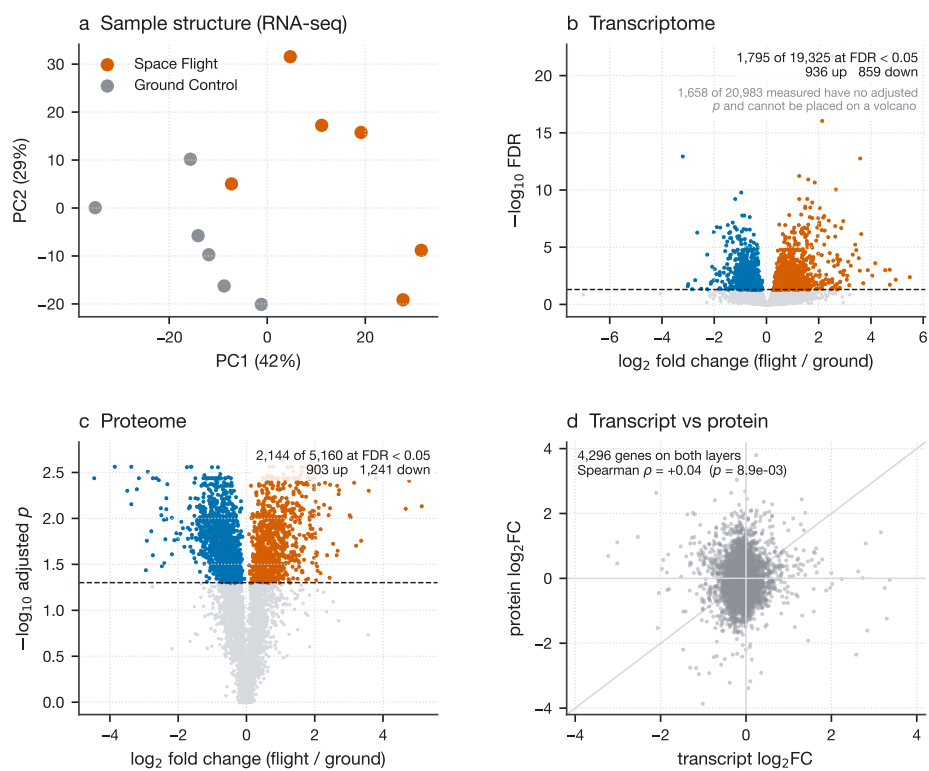


Fig. 3 The metabolite layer had to be predicted. Every value shown is model output.

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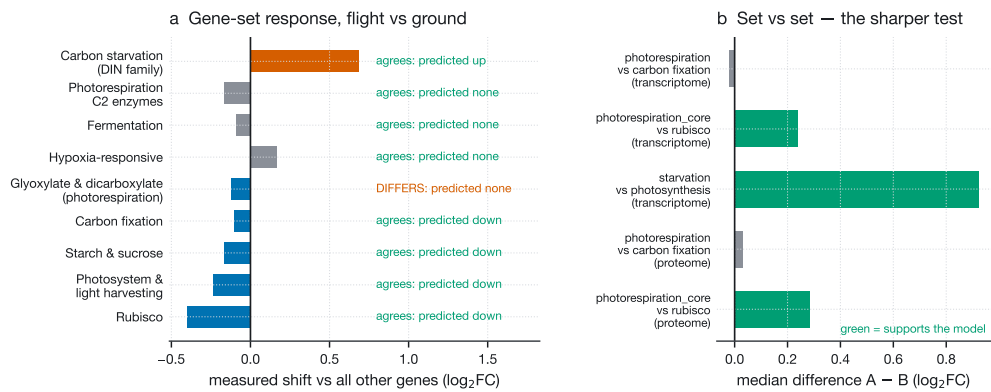
OSD-522: Arabidopsis shoots, 6 flight vs 6 ground, in BRIC-LED hardware



Transcriptome computed here with PyDESeq2; proteome ratios as deposited by the original study. Both measured.

Fig. 4 OSD-522: *Arabidopsis* shoots, six flight against six ground.

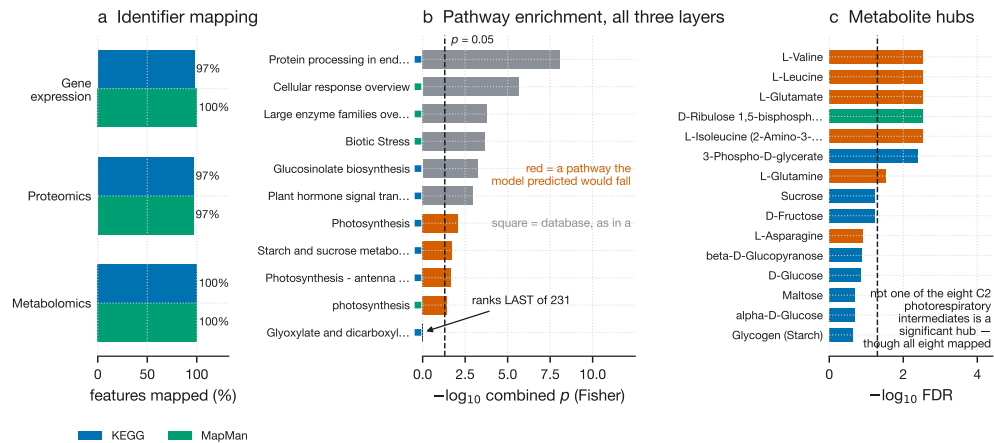
A model that never saw the data called eight of nine gene sets



Predictions fixed before any omics were downloaded; gene sets from KEGG pathway membership and descriptions.

Fig. 5 The model's blind prediction against the measurement.

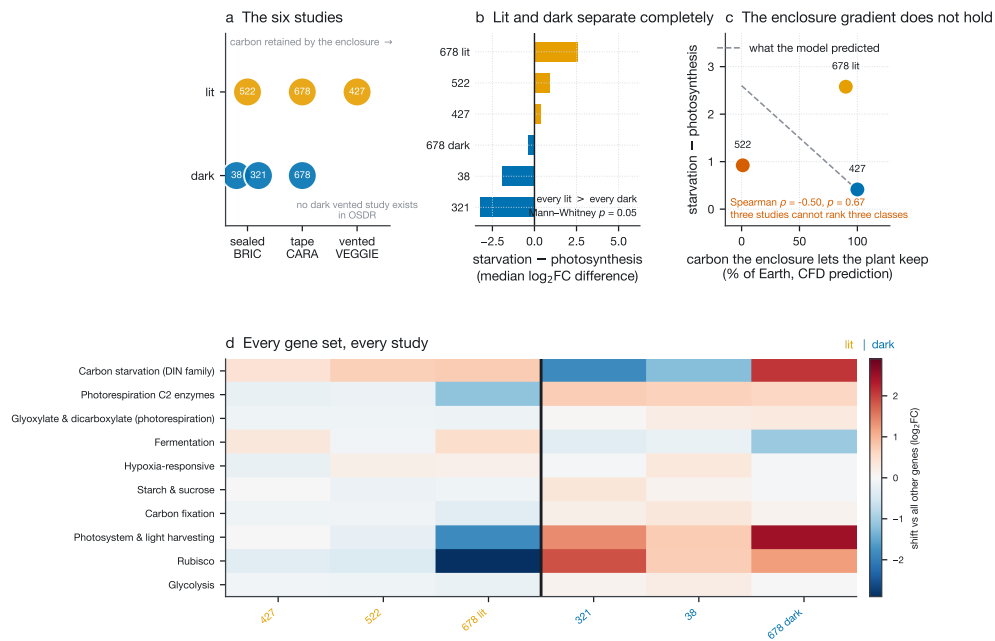
PaintOmics: photosynthesis enriched, photorespiration last of 231



Job m1z16Qg3DK, organism ath, KEGG + MapMan, AI interpretation off. Hub analysis counts REAL differentially expressed genes.

Fig. 6 Pathway integration across all three layers. In **b** a square marks each pathway's database, in the colours of **a**: KEGG's *Photosynthesis* and MapMan's *photosynthesis* are different maps that differ only in capitalisation, and both are significant.

Illumination, not sealing, is what orders the spaceflight response



Six matched flight-vs-ground contrasts from NASA OSDR; gene sets from KEGG. Study selection in data/study_registry.tsv.

Fig. 7 Illumination separates the six studies; the enclosure gradient does not.

Designing the experiment that could tell hardware from gravity

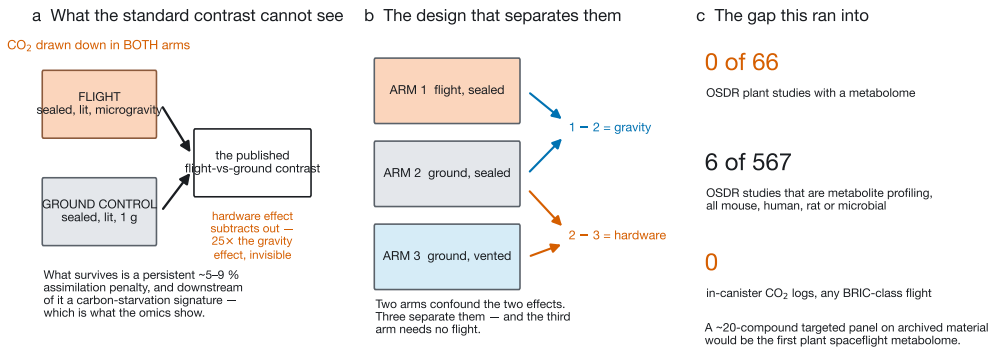


Fig. 8 Designing the experiment that could tell hardware from gravity.